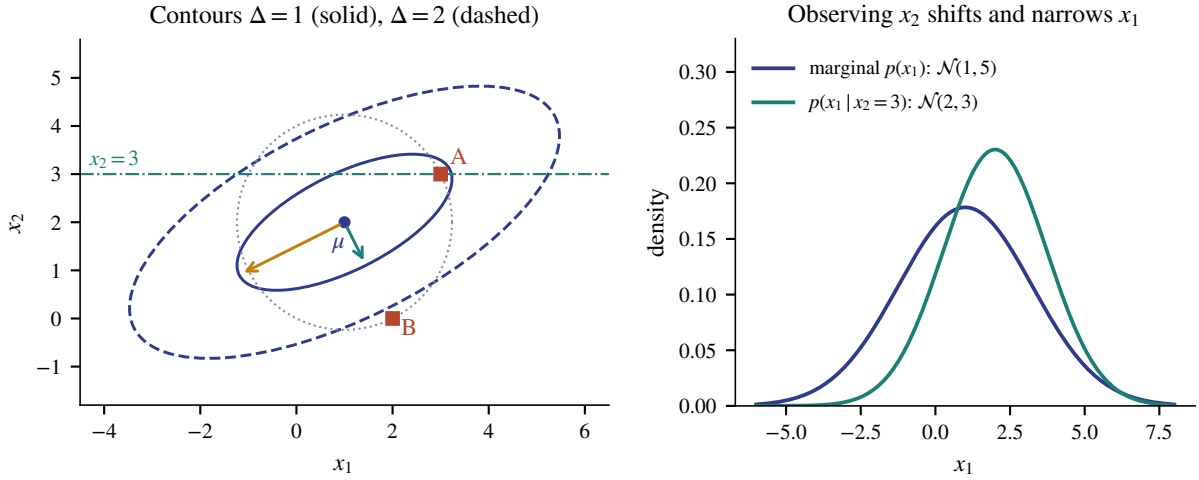
Contours Δ = 1 (solid), Δ = 2 (dashed)
$x_2 = 3$
A
B
$\mu$
$x_1$
$x_2$
Observing $x_2$ shifts and narrows $x_1$
marginal $p(x_1)$: $\mathcal{N}(1, 5)$
$p(x_1 \mid x_2 = 3)$: $\mathcal{N}(2, 3)$
density
$x_1$